

Class 1 Reserved-Slot Strategies: Day-Level Formulation

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1 Introduction

1.1 Purpose and Reader Map

This note is a companion to `first_two_class_simulation_note.v2.tex`. It keeps the same day-level notation, the same patient behavior objects, and the same metric interpretation, but it adds a strict booking strategy in which Class 1 has protected daily capacity.

Core Policy Question

The pooled FCFS baseline asks which patient arrives first. The reserved-slot strategy asks whether some daily capacity should be protected for Class 1 before the two classes compete for the remaining slots.

The two strategies compared in this note are:

- **Pooled FCFS:** every class competes for the same S daily slots.
- **Strict Class 1 reservation:** $Q = 10$ slots per day are reserved for Class 1 only; unused reserved slots stay empty.

Future Focus: Soft Reservation

A soft reservation or Class-1-priority backfill policy remains a possible future extension. It is not part of the current code or notebook exploration.

1.1.1 What Changes from the FCFS Note

Object	FCFS baseline	Reserved-slot replacement
Daily capacity	one pooled capacity S	protected capacity Q plus general capacity $S - Q$
Calendar state	$X_{i,r}^D, N_r^D$	same $X_{i,r}^D$, plus audit-layer pool counts $X_{i,r,\text{res}}^D$ and $X_{i,r,\text{gen}}^D$
Offer rule	earliest residual day with $N_r^D < S$	earliest residual day with eligible pool capacity
No-offer logic	no pooled slot is available	no eligible pool is available for that class and policy
Service, cancellation, no-show	unchanged	unchanged

2 Model Formulation

2.1 Notation Inherited from v2

The simulation clock is still the day D . The basic objects are unchanged:

- S : total appointment slots available per day.
- H : rolling booking horizon in days.
- $\mathcal{I} = \{1, 2\}$: patient classes.
- $r \in \{0, 1, \dots, H - 1\}$: residual delay, where $r = 0$ means appointments scheduled for today.
- τ : original offered delay accepted by a booked patient.
- $X_{i,r}^D$: number of class- i patients scheduled for day $D + r$.
- $N_r^D = \sum_{i \in \mathcal{I}} X_{i,r}^D$: total scheduled patients on residual day r .

The same residual-delay/original-delay distinction remains:

Residual Delay vs. Original Delay

Residual delay r is part of the rolling calendar state and decreases when the day advances. Original delay τ is stored with the booking because balking and no-show behavior use the delay the patient was originally offered.

To make the policy formulas match the current implementation and remain compatible with the v2 notation, define the bookable residual-day set

$$\mathcal{R} \subseteq \{0, 1, \dots, H - 1\}.$$

In the current Python engine, same-day booking is allowed, so

$$\mathcal{R} = \{0, 1, \dots, H - 1\}.$$

If a future-only convention is desired, the same formulas apply after replacing $\mathcal{R}$ with $\{1, \dots, H - 1\}$.

2.2 Reservation Parameters

Let the protected class be

$$i^* = 1,$$

and let

$$Q = 10$$

be the number of Class 1 reserved slots per day. Since the baseline capacity is $S = 32$, the general pool has

$$S - Q = 22$$

slots per day.

The total day-level state $X_{i,r}^D$ is still the main state, but the reserved policies require an audit-layer pool label. Write

$$X_{i,r,\text{res}}^D = \text{number of class-}i \text{ patients scheduled in reserved slots on day } D + r,$$

and

$$X_{i,r,\text{gen}}^D = \text{number of class-}i \text{ patients scheduled in general slots on day } D + r.$$

Then

$$X_{i,r}^D = X_{i,r,\text{res}}^D + X_{i,r,\text{gen}}^D.$$

Define the pool occupancies

$$R_r^D = \sum_{i \in \mathcal{I}} X_{i,r,\text{res}}^D, \quad G_r^D = \sum_{i \in \mathcal{I}} X_{i,r,\text{gen}}^D,$$

so that

$$N_r^D = R_r^D + G_r^D.$$

The capacity constraints are

$$R_r^D \leq Q, \quad G_r^D \leq S - Q, \quad N_r^D \leq S.$$

Daily Capacity Split for the Implemented Strategy

Pool	Capacity	Eligibility
Reserved pool	$Q = 10$	Class 1 only
General pool	$S - Q = 22$	Class 1 overflow and Class 2
Total day	$S = 32$	all scheduled patients, regardless of pool label

2.3 Patient Behavior is Unchanged

The class behavior object remains

$$C_i = \{\lambda_i, b_i(\tau), \phi_i, \xi_i(\tau)\}.$$

The reservation policy does not change arrival rates, balking, cancellation, or no-show behavior. They only change which residual day and which capacity pool an arriving patient is offered.

What the Reservation Strategy Does Not Change

For a class- i patient offered delay τ , the balking probability remains $b_i(\tau)$. For a booked class- i patient scheduled in the future, the daily cancellation probability remains ϕ_i . At service time, the no-show probability remains $\xi_i(\tau)$.

3 Booking Rules

3.1 Pooled FCFS Baseline

The pooled FCFS baseline searches the residual days in order and offers the first day with open total capacity:

$$\tau_{\text{FCFS}} = \min \{r \in \mathcal{R} : N_r^D < S\}.$$

If the set is empty, the patient receives no offer.

If the patient accepts, the simulator increments

$$X_{i,\tau_{\text{FCFS}}}^D \leftarrow X_{i,\tau_{\text{FCFS}}}^D + 1$$

and stores the audit record

$$(i, \tau_{\text{FCFS}}).$$

FCFS in One Line

Under pooled FCFS, class labels do not affect the offer rule. They affect behavior after the offer through $b_i(\tau)$, ϕ_i , and $\xi_i(\tau)$, but not the capacity search itself.

3.2 Strict Class 1 Reservation

Under strict reservation, Class 1 has first access to the reserved pool on each day. If the reserved pool is already full on the earliest eligible day, Class 1 can compete for general slots. Class 2 can use only the general pool. Unused reserved slots stay empty and are carried forward as protected future capacity until the appointment day arrives.

3.2.1 Class 1 Offer Rule

For a Class 1 arrival, the strict policy searches residual days in increasing r . On each residual day, it checks the reserved pool first and the general pool second:

$$\tau_{1,\text{strict}} = \min \{r \in \mathcal{R} : R_r^D < Q \text{ or } G_r^D < S - Q\}.$$

The pool label is

$$z_{1,\text{strict}}(r) = \begin{cases} \text{res}, & R_r^D < Q, \\ \text{gen}, & R_r^D = Q \text{ and } G_r^D < S - Q. \end{cases}$$

If both pools are unavailable for every $r \in \mathcal{R}$, the Class 1 patient receives no offer. If the patient accepts, the corresponding pool count is incremented:

$$X_{1,\tau,\text{res}}^D \leftarrow X_{1,\tau,\text{res}}^D + 1 \quad \text{or} \quad X_{1,\tau,\text{gen}}^D \leftarrow X_{1,\tau,\text{gen}}^D + 1.$$

3.2.2 Class 2 Offer Rule

For a Class 2 arrival under strict reservation, the reserved pool is invisible:

$$\tau_{2,\text{strict}} = \min \{r \in \mathcal{R} : G_r^D < S - Q\}.$$

If the set is empty, the Class 2 patient receives no offer even if some reserved capacity remains unused:

$$R_r^D < Q \quad \text{does not help Class 2 under strict reservation.}$$

If the patient accepts, the simulator increments

$$X_{2,\tau,\text{gen}}^D \leftarrow X_{2,\tau,\text{gen}}^D + 1.$$

Strict Reservation Interpretation

Strict reservation protects Class 1 access at the cost of possible empty protected capacity. It is not a utilization-maximizing policy; it is an access-protection policy.

4 Visual Calendar Logic

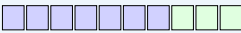
4.1 One Residual Day

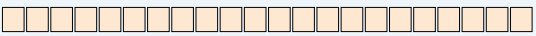
The reservation logic is easiest to see on a single residual day r . The day has $Q = 10$ protected slots and $S - Q = 22$ general slots.

Pool Occupancy on One Residual Day

Suppose $S = 32$, $Q = 10$, and the post-cancellation state for one residual day is

$$R_r^D = 7, \quad G_r^D = 22, \quad N_r^D = 29.$$

reserved $Q = 10$  $R_r^D = 7$, three protected slots open

general $S - Q = 22$  $G_r^D = 22$, no general slots open

Under strict reservation, a Class 1 patient can be offered this day, but a Class 2 patient cannot use the open protected slots.

4.2 Residual-Day Search

Comparing the Earliest Feasible Day

Take $H = 4$, $S = 32$, $Q = 10$, and the post-cancellation state below.

Count	day D $r = 0$	day $D + 1$ $r = 1$	day $D + 2$ $r = 2$	day $D + 3$ $r = 3$
R_r^D	10	8	10	6
G_r^D	22	22	20	13
N_r^D	32	30	30	19

The implied offers are:

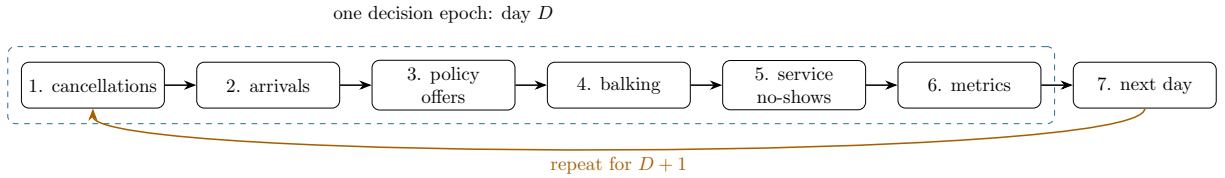
Policy and class	Earliest offer	Reason
Pooled FCFS, either class	$r = 1$	$N_1^D = 30 < 32$
Strict reservation, Class 1	$r = 1$, reserved	$R_1^D = 8 < 10$
Strict reservation, Class 2	$r = 2$, general	$G_1^D = 22$ is full, but $G_2^D = 20 < 22$

5 Daily Simulation Mechanics

5.1 Unchanged Daily Order

The daily event order remains the same as in the v2 note:

1. **Compute cancellations for the day.** Future appointments independently cancel with probability ϕ_i .
2. **Simulate daily arrivals.** $M_i^D \sim \text{Poisson}(\lambda_i)$.
3. **Generate a randomized within-day arrival order.**
4. **Apply the configured offer rule.** This is the only step changed by the reserved-slot strategy.
5. **Simulate balking.** A class- i patient offered delay τ accepts with probability $1 - b_i(\tau)$.
6. **Simulate no-shows and service for day D .**
7. **Compute metrics and advance to day $D + 1$.**



5.2 Strict Reservation Day

Under strict reservation, the randomized arrival order is processed once. The difference from pooled FCFS is that the feasible capacity set depends on the patient's class:

$$\mathcal{F}_{1,\text{strict}}^D = \{(r, \text{res}) : r \in \mathcal{R}, R_r^D < Q\} \cup \{(r, \text{gen}) : r \in \mathcal{R}, G_r^D < S - Q\},$$

where reserved capacity is checked before general capacity on the same residual day. For Class 2,

$$\mathcal{F}_{2,\text{strict}}^D = \{(r, \text{gen}) : r \in \mathcal{R}, G_r^D < S - Q\}.$$

If $\mathcal{F}_{i,\text{strict}}^D$ is empty, that patient receives no offer. The simulator then continues to later arrivals because a later Class 1 arrival may still have protected capacity even when a Class 2 arrival has no general capacity.

6 Comparison to Pooled FCFS

6.1 Policy Effects

Feature	Pooled FCFS	Strict Class 1 reservation
Capacity structure	all S slots pooled	Q protected, $S - Q$ general
Class 1 access	competes only by arrival order	protected first, then general overflow
Class 2 access	can use any open slot	can use only general slots
Unused reserved slots	not applicable	may remain empty
Utilization tendency	highest when demand is sufficient and no-shows are low	can fall if Class 1 demand is not enough to use Q
Equity/access tendency	neutral in the offer rule; differences come from behavior	shifts access toward Class 1

6.2 Expected Metric Patterns

The main metrics are unchanged:

$$\rho^{\text{attended}} = \frac{\text{served slots}}{S \times \text{measured days}}, \quad \rho^{\text{booked}} = \frac{\text{booked slots on service days}}{S \times \text{measured days}}, \quad \text{served rate}_i = \frac{\text{served}_i}{\text{arrivals}_i}.$$

ρ^{attended} is completed-visit utilization. For the reservation analysis, ρ^{booked} is also useful because it shows whether protected slots were actually occupied on the schedule. No-shows count as booked slots; appointments that cancel early enough to free the service-day slot do not.

The reservation strategy mainly affects the path to those metrics:

$$\text{offer rule} \rightarrow \tau \rightarrow b_i(\tau) \rightarrow \text{booked patients} \rightarrow \phi_i, \xi_i(\tau) \rightarrow \text{served patients}.$$

Interpretation of the Comparison

Strict reservation can improve Class 1 access while lowering booked-slot utilization if protected slots are not filled. It can also lower attended utilization if booked slots turn into no-shows.

6.3 Objective-Function Analysis

The metrics above describe what happened in one simulation run. The objective-function analysis adds one more step: it turns each FCFS run and each strict-reservation run into a single comparable

score. This is useful because strict reservation can improve Class 1 outcomes while also leaving some protected capacity unused. In this section the main capacity score is booked-slot utilization, because the policy question is whether protected slots are actually occupied. Completed-visit utilization remains a related diagnostic for no-show losses. A single metric is therefore not enough to decide whether a given Q is attractive.

The objective functions are not a new booking policy. The booking policies remain pooled FCFS and strict Class 1 reservation. The objective functions are only used after the simulation to compare the resulting outcomes.

6.3.1 Policy Parameter

The main policy parameter is the protected daily capacity

$$Q = \text{number of Class 1 reserved slots per day.}$$

In the visual examples above, $Q = 10$ and $S = 32$ are fixed to make the mechanics concrete. In the objective-function notebooks, Q is swept across a range of values and compared against the matched FCFS baseline, where $Q = 0$.

It is helpful to scale the cost of reservation by the protected share of the day:

$$\frac{Q}{S}.$$

This treats one protected slot as more expensive when the clinic has fewer total daily slots.

6.3.2 Initial Analysis Grid

The first objective-function notebooks center the sweep around the realistic demand scale rather than the larger visual example in this note. The baseline analysis uses total daily demand $\lambda \approx 25$, Class 1 share $p = 0.58$, and daily capacity $S = 20$. The main Q sweep is

$$Q \in \{0, 1, 2, 3, 4, 5, 6, 8, 10, 12\}.$$

Demand and mix are varied around that center:

$$\lambda \in \{15, 20, 25, 30, 35, 40, 50\}, \quad p \in \{0.40, 0.50, 0.58, 0.65, 0.75\}.$$

The initial weight and penalty values are

$$w_1 \in \{1, 1.25, 1.5, 2, 3\}, \quad c \in \{0, 0.02, 0.05, 0.10\}, \quad \gamma \in \{0, 0.02, 0.05, 0.10\},$$

with $w_2 = 1$. The headline setting is $w_1 = 1.5$, $c = 0.05$, and $\gamma = 0.05$. These values are not claimed to be final; they are starting values for finding which parts of the demand and behavior space make strict reservation look better than FCFS.

6.3.3 Weighted Booked Slot Use

For class-specific booked slot use, define each class contribution as

$$\rho_i^{\text{booked}} = \frac{\text{served}_i + \text{no show}_i}{S \times \text{measured days}}.$$

The weighted booked-slot objective is

$$U_{\text{booked}}(w_1, w_2) = w_1 \rho_1^{\text{booked}} + w_2 \rho_2^{\text{booked}}.$$

Class 2 is kept at $w_2 = 1$. Increasing w_1 makes Class 1 booked slots count more in the comparison. When $w_1 = w_2 = 1$, this reduces to total booked-slot utilization ρ^{booked} .

The completed-visit version is still available as a diagnostic:

$$\rho_i^{\text{attended}} = \frac{\text{served}_i}{S \times \text{measured days}}.$$

This separates empty protected slots from booked slots that later no-show.

6.3.4 Weighted Served Rate

The class served-rate metric from the previous subsection is

$$\text{served rate}_i = \frac{\text{served}_i}{\text{arrivals}_i}.$$

The weighted served-rate objective compares weighted served patients to weighted arrivals:

$$U_{\text{served}}(w_1, w_2) = \frac{w_1 \text{served}_1 + w_2 \text{served}_2}{w_1 \text{arrivals}_1 + w_2 \text{arrivals}_2}.$$

This objective is bounded between 0 and 1 before any penalty is applied. It answers a direct access question: after giving Class 1 a higher weight, what share of weighted demand was served?

6.3.5 Slot-Cost Penalty

Strict reservation protects access, but protected slots can be left empty if Class 1 demand is not large enough or if Class 1 behavior leads to cancellations, no-shows, or balking. To reflect that tradeoff, the net booked-slot objective subtracts a cost for configured protected capacity:

$$U_{\text{net booked}}(w_1, w_2, c) = U_{\text{booked}}(w_1, w_2) - c \frac{Q}{S}.$$

Here c controls how expensive protected capacity is. A larger c makes high- Q policies harder to justify unless they create enough weighted booked-slot benefit. The notebooks also keep the served-rate version $U_{\text{served}}(w_1, w_2) - cQ/S$ as an access diagnostic.

6.3.6 Waiting-Time Penalty

Because the simulator stores the originally offered delay τ , a waiting-time version can also be evaluated. For offered patients, define the weighted average offered delay

$$\bar{\tau}_w = \frac{w_1 \sum_{\text{offered class 1}} \tau + w_2 \sum_{\text{offered class 2}} \tau}{w_1 \text{offered}_1 + w_2 \text{offered}_2}.$$

The wait-adjusted objective is

$$U_{\text{wait}}(w_1, w_2, c, \gamma) = U_{\text{booked}}(w_1, w_2) - c \frac{Q}{S} - \gamma \frac{\bar{\tau}_w}{H}.$$

The term $\bar{\tau}_w/H$ normalizes offered delay by the booking horizon. The parameter γ controls how strongly longer offered waits should lower the score.

6.3.7 Basic Requirements

Some comparisons are easier to read as a requirements check rather than a single score. The notebooks therefore also record whether a run satisfies

$$\rho^{\text{booked}} \geq \rho_{\min}, \quad \min_i \text{ served rate}_i \geq x, \quad \left| \rho_1^{\text{booked}} - \rho_2^{\text{booked}} \right| \leq g.$$

Here $\rho_{\min}$ is the minimum booked-slot utilization level, x is the minimum served-rate floor for each class, and g is the largest allowed gap between the two class-specific booked-slot utilization contributions.

6.3.8 Matched Comparison to FCFS

Each strict-reservation run is compared to a pooled FCFS run with the same scenario and seed. For a scenario a , seed j , and objective U , the paired difference is

$$\Delta U(Q; a, j) = U_{\text{strict}}(Q; a, j) - U_{\text{FCFS}}(a, j).$$

The sweep averages this paired difference over J seeds:

$$\overline{\Delta U}(Q) = \frac{1}{J} \sum_{j=1}^J \Delta U(Q; a, j).$$

The best protected capacity for a scenario is the Q with the highest average paired difference:

$$Q^*(a) = \arg \max_Q \overline{\Delta U}(Q; a).$$

The notebooks label a setting as a clear win when the average paired difference is positive and the lower end of the 95 percent interval is also positive. If the average is positive but the interval crosses zero, the setting is a possible win. Otherwise it is treated as a loss relative to FCFS.

How to Read the Objective Analysis

The goal is not to find one universal Q . The goal is to find when strict reservation beats FCFS for a stated priority weight, a stated cost of protected capacity, and a stated demand and behavior scenario.

6.4 No-Offer Interpretation

The no-offer event has a slightly different meaning across the strategies:

Policy	No-offer meaning
Pooled FCFS	no day in $\mathcal{R}$ satisfies $N_r^D < S$
Strict reservation, Class 1	no day has either $R_r^D < Q$ or $G_r^D < S - Q$
Strict reservation, Class 2	no day has $G_r^D < S - Q$, even if $R_r^D < Q$ somewhere

7 Implementation Mapping

7.1 Config Objects

The new implementation fields map directly to the notation:

Code field	Notation	Meaning
slots_per_day	S	total daily capacity
reserved_class_id	i^*	protected class; here $i^* = 1$
reserved_slots_per_day	Q	protected daily capacity; here $Q = 10$
Booking.reserved_slot	pool label z	true means the appointment was placed in the reserved pool

7.2 Code Path

The policy logic appears in the booking step:

- `find_earliest_open_day` computes the earliest feasible (r, z) pair.
- `process_daily_arrivals` applies the one-pass arrival order.
- For Class 1, `find_earliest_open_day` searches each residual day as reserved then general.
- For Class 2, `find_earliest_open_day` searches general capacity only.

Review Against the Implemented Reserved-Slot Model

- The original day-level state notation $X_{i,r}^D$, N_r^D , r , τ , S , and H is unchanged.
- The new audit-layer pool counts R_r^D and G_r^D split the same daily capacity into reserved and general pools.
- $Q = 10$ protected slots are reserved for Class 1 out of $S = 32$.
- In strict reservation, unused protected slots stay unavailable to Class 2.
- Cancellation, no-show, service, and metric definitions remain the same as in the FCFS v2 note.

8 Conclusion

The reservation strategy is not a separate patient behavior model. It is a change to the offer rule inside the same day-level appointment simulator. Pooled FCFS uses one capacity constraint

$$N_r^D < S.$$

Strict Class 1 reservation replaces that with class-dependent eligibility over

$$R_r^D < Q \quad \text{and} \quad G_r^D < S - Q.$$

The comparison should therefore be read as an access/utilization tradeoff:

$$\text{more Class 1 protection} \iff \text{less automatic pooling of daily capacity.}$$